

Other Outcomes: Ordinal, Nominal, Count, Duration

Francisco Villamil

Applied Quantitative Methods II
MA in Social Sciences, Spring 2026

Logistics

- Presentations (April 16th)
 - **5-minute presentations**, blocks of ≈ 5 , random order, 15-min feedback after each block
<https://franvillamil.github.io/AQM2/logistics.html>
 - Send me **presentations** by email **in advance**
3 slides max!: e.g. what, how, problems
- Rest of the course
 - Computing session (April 23rd)
 - Final session (April 30th), **exam** + review

Today's goals

- Understand the GLM framework as a unifying structure
- Model ordinal outcomes with ordered logit
- Model nominal (multi-category) outcomes with multinomial logit
- Model count data with Poisson and negative binomial regression
- Model time-to-event data with survival analysis and the Cox model
- Know which model to reach for and why

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

One framework, many models

Generalized Linear Models (GLMs) share three components:

Linear predictor

$$\eta_i = \mathbf{x}_i\boldsymbol{\beta}$$

A weighted sum of
covariates

Link function

$$g(\mu_i) = \eta_i$$

Connects mean to
predictor

Distribution

$$Y_i \sim F(\mu_i)$$

Data-generating process

- OLS is a GLM: identity link + normal distribution

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Data-generating process

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- Logit is a GLM: logit link + Bernoulli distribution
- Choosing a model = choosing a link + distribution

Matching outcome type to model

Outcome type	Distribution	Link	Model / Function
Continuous	Normal	Identity	OLS (<code>lm()</code>)
Binary (0/1)	Bernoulli	Logit	Logit (<code>glm(...,binomial)</code>)
Ordinal	Prop. odds	Cumul. logit	Ordered logit (<code>polr()</code>)
Nominal (>2)	Multinomial	Log-odds	Multinom. logit (<code>multinom()</code>)
Count (≥ 0)	Poisson	Log	Poisson (<code>glm(...,poisson)</code>)
Count (overdisp)	Neg. Binomial	Log	NegBin (<code>glm.nb()</code>)
Duration (time)	—	—	Cox PH (<code>coxph()</code>)

First question: what is the data-generating process for your outcome?

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Ordinal outcomes:

When categories have an order but not equal spacing

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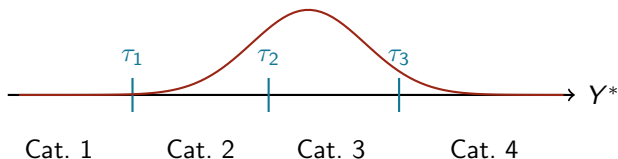
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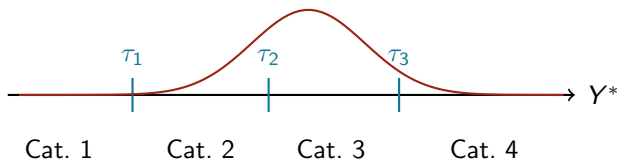
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- OLS treats them as equally spaced — often wrong

The latent variable idea



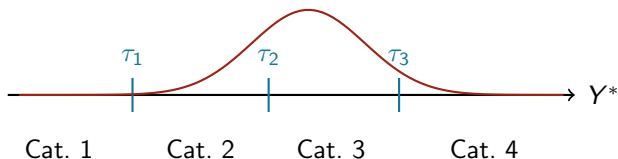
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- We observe category j when $\tau_{j-1} < Y^* \leq \tau_j$
- Thresholds τ_j are **estimated** from the data

Ordered logit: the model

Cumulative link model (proportional odds)

$$\Pr(Y_i \leq j \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\tau_j - \mathbf{x}_i\beta))}$$

- One set of β for all thresholds (**proportional odds assumption**)

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- $\beta_k > 0$ shifts Y^* upward $\rightarrow$ **higher** categories become more likely
- $\Pr(Y_i = j) = \Pr(Y_i \leq j) - \Pr(Y_i \leq j - 1)$: probability for each category by differencing cumulative probabilities

Ordered logit in R

```
library(MASS); library(marginaleffects)
m_ord = polr(factor(satisfaction) ~ age + educ,
              data = mydata, Hess = TRUE)
avg_slopes(m_ord) # AMEs per predictor per category
```

- Hess = TRUE: stores the Hessian — required for SEs

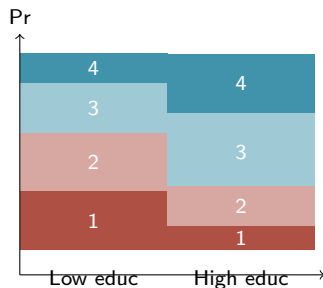
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- **Sign warning:** raw polr() coefficients use a reversed sign convention — always interpret via avg_slopes()

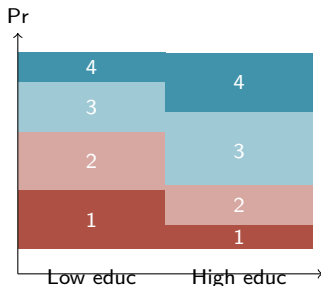
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive



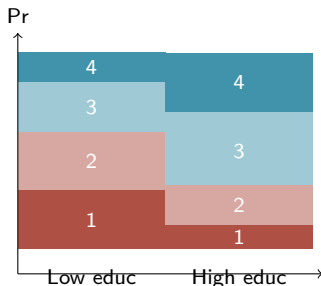
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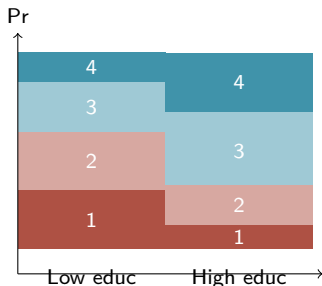
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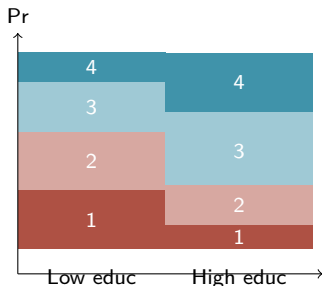
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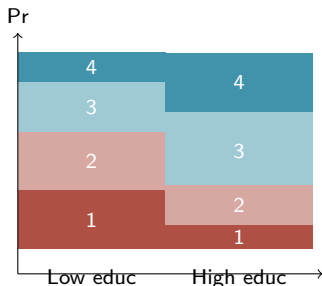
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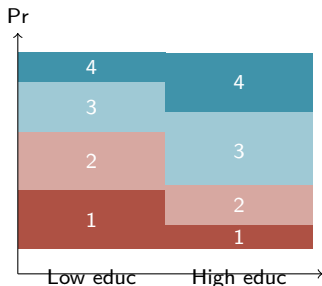
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 - How does $\Pr(Y = j)$ change across X ?



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 - How does $\Pr(Y = j)$ change across X ?
 - Use `plot_predictions(m_ord, condition = "educ")`



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Nominal outcomes:

Multiple unordered categories — vote choice, transport mode,
conflict type

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 - Occupation category (manager, professional, manual, service)
- Ordered logit is wrong here — it would impose an ordering
- We need a model that treats all categories symmetrically

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\beta'_j \mathbf{x}_i)}{\sum_{k=1}^J \exp(\beta'_k \mathbf{x}_i)}$$

- J categories $\rightarrow J - 1$ sets of coefficients (one category is the **reference**)

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- All predicted probabilities sum to 1: $\sum_{j=1}^J \Pr(Y_i = j) = 1$

Multinomial logit in R

```
library(nnet)

m_mnom = multinom(vote ~ age + ideology + income,
  data = beps)

# AMEs (one per predictor per category)
avg_slopes(m_mnom)
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- First category = reference; coefficients = log-odds vs. reference

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- Change reference: `relevel(factor(vote), ref = "Labour")`

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- Change reference: `relevel(factor(vote), ref = "Labour")`
- Interpret via `avg_slopes()` or `plot_predictions()`

The IIA assumption

Independence of Irrelevant Alternatives (IIA): the odds ratio between any two alternatives is **unaffected** by the presence or absence of other alternatives.

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp((\beta_j - \beta_k)' \mathbf{x})$$

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You are modeling vote choice in a Spanish election with five parties.

Two parties are both center-left. Should you worry about the IIA assumption?

What would you do?

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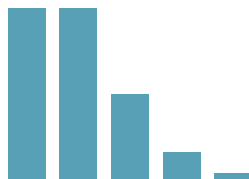
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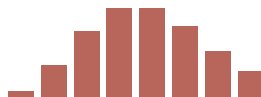
Non-negative integers — conflicts, protests, bills, articles

Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...



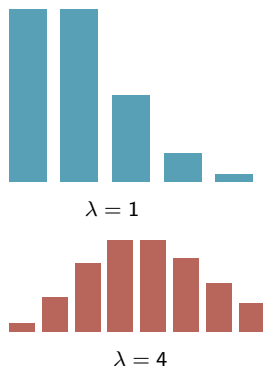
$\lambda = 1$



$\lambda = 4$

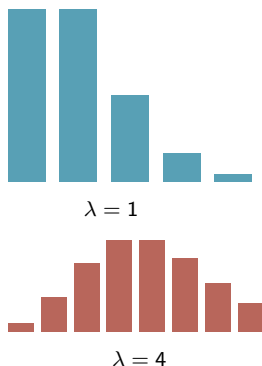
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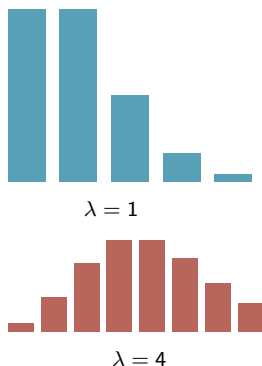
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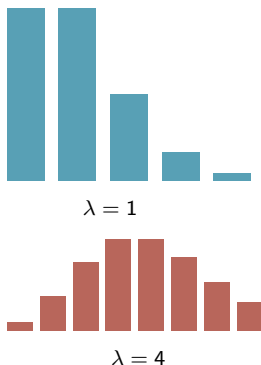
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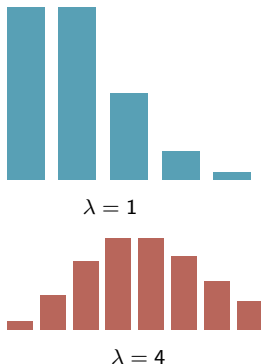
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- **Why not OLS?**
 - Counts ≥ 0 : OLS can predict negative values
 - Variance increases with the mean
 - Strongly right-skewed; often many zeros



The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i\boldsymbol{\beta})$$

$$\Leftrightarrow \log(\mu_i) = \mathbf{x}_i\boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)

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 - $\beta_k = 0.3 \Rightarrow \exp(0.3) \approx 1.35$: 35% increase in expected count
- Poisson assumption: $\text{Var}(Y_i) = E[Y_i]$ (mean = variance)

Poisson in R and interpretation

```
# Fit Poisson model  
m_pois = glm(n_conflicts ~ gdp_pc + pop_l,  
             family = "poisson", data = mydata)  
  
summary(m_pois)  
  
exp(coef(m_pois)) # incidence rate ratios
```

Raw coefficient

$$\hat{\beta}_{\text{gdp_pc}} = -0.24$$

Hard to interpret directly

Exponentiated

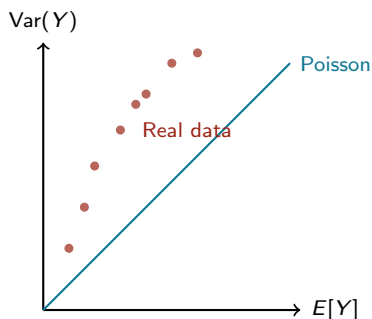
$$\exp(-0.24) \approx 0.79$$

→ 21% fewer conflicts per unit GDP

Also: `avg_slopes(m_pois)` gives AMEs on count scale

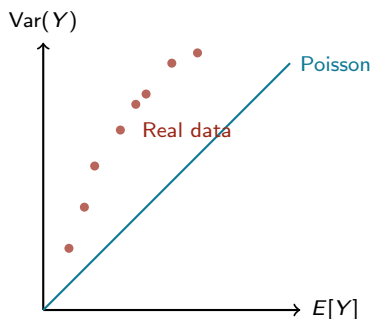
The overdispersion problem

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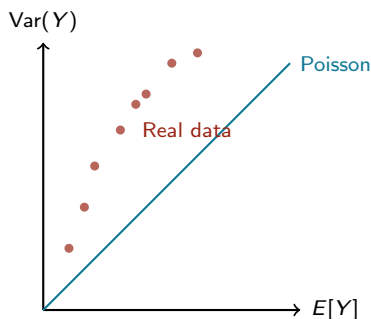
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- Real data almost always:
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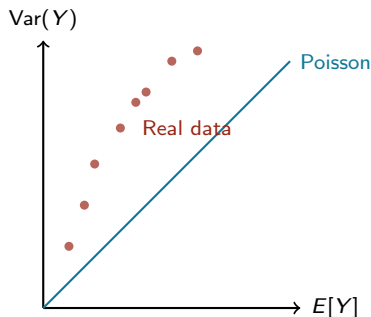
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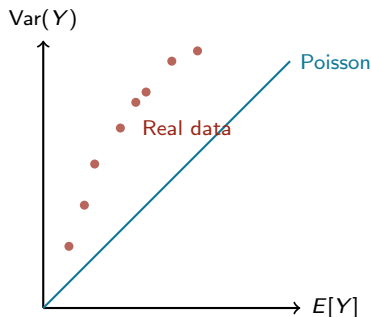
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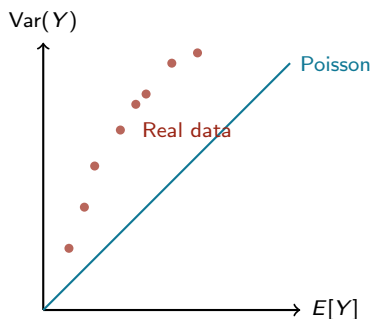
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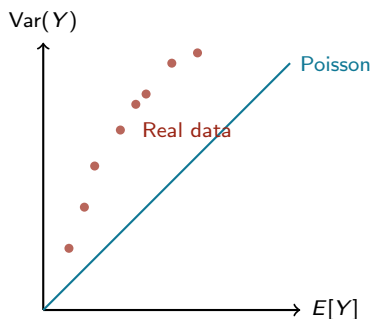
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- Quick check:
Residual deviance $\gg$ df



Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data

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- In R: `MASS::glm.nb(y ~ x1 + x2)`

Count models: a summary

Model		R function	When to use
Poisson		<code>glm(...,poisson)</code>	Equidispersed counts
Negative binomial		<code>MASS::glm.nb()</code>	Overdispersed counts
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- **Default:** start Poisson, test overdispersion, switch to NegBin if needed
- Zero-inflation: two-process model (structural vs. count zeros)
- All share the log link: $\exp(\hat{\beta}) = \text{incidence rate ratio (IRR)}$

Discussion

You model the number of conflict events per country-year.

*After fitting a Poisson model, you check the output: Residual
deviance: 847 on 142 df.*

What does this tell you? What would you do next?

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Duration analysis:

Time until an event — war onset, cabinet survival, regime
collapse

Time-to-event outcomes

- Outcomes that record **how long until something happens**

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- Why not OLS on duration?

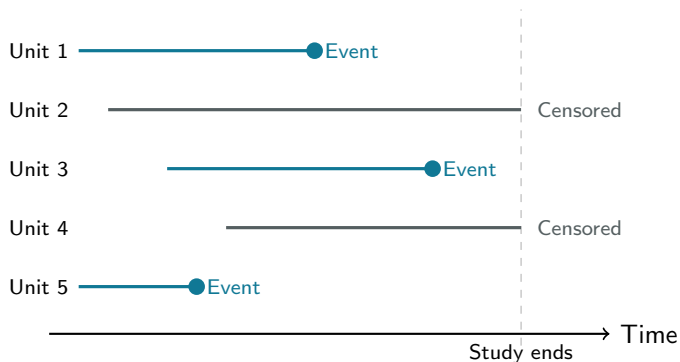
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 - **Censoring**: many observations end without the event occurring

The censoring problem



Censored units: event did not occur during observation window

Discussion

A researcher studies how long democracies survive.

She drops all country-spells that are still ongoing at the end of the dataset, keeping only countries where a democratic breakdown was observed.

What is wrong with this decision?

Key concepts: survival and hazard

Survival function $S(t)$

$$S(t) = \Pr(T > t)$$

Probability of surviving past time t

$$S(0) = 1, S(\infty) = 0$$

Monotonically non-increasing

- High $h(t)$: event is likely soon

Hazard function $h(t)$

$$h(t) = \frac{f(t)}{S(t)}$$

Instantaneous rate of event at t ,
conditional on survival to t

“Danger per unit time at time t ”

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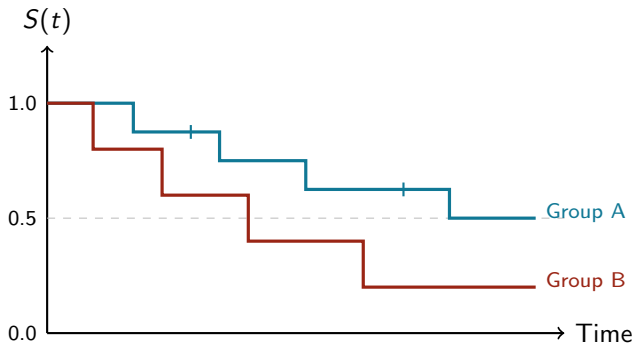
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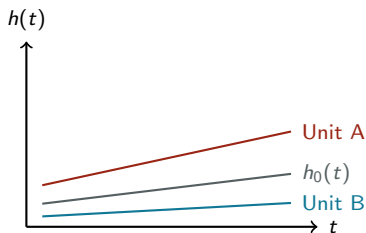
- High $h(t)$: event is likely soon
- $S(t)$ and $h(t)$ are two views of the same object
- Covariates shift $h(t)$ — that is what regression models

Kaplan-Meier: visualizing survival



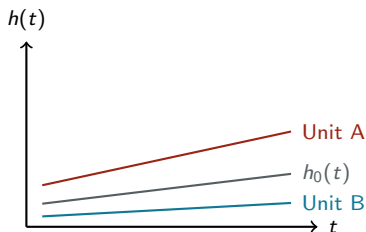
Kaplan-Meier estimator: non-parametric, no covariates, visual comparison of groups

The proportional hazards idea



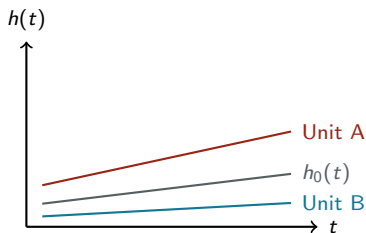
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- The ratio $h_i(t)/h_j(t)$ is **constant over time** — curves never cross
- Violation: if the gap between lines closes or reverses over time $\Rightarrow$ PH assumption fails

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i\boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)

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 - $\exp(\beta_k) < 1$: lower hazard (event happens later)

Cox model in R

```
library(survival); library(survminer)

# Kaplan-Meier estimator
km_fit = survfit(Surv(time, event) ~ group, data = d)
ggsurvplot(km_fit, conf.int = TRUE)

# Cox PH model
cox_fit = coxph(Surv(time, event) ~ x1 + x2, data = d)
summary(cox_fit) # shows hazard ratios
exp(coef(cox_fit))
```

Interpreting the Cox model

	Coef	Exp(coef)	SE	p-value
gdp_pc	-0.31	0.73	0.08	< 0.001
military	0.64	1.90	0.15	0.001
log_pop	0.18	1.20	0.10	0.07

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Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Pulling it together:

Which model for which data?

Choosing the right model

If your outcome is...	Use	Key check
Continuous	OLS	Linearity, homoskedasticity
Binary (0/1)	Logit / LPM	Rare events → prefer logit
Ordered categories	Ordered logit	Test PO assumption (Brant)
Unordered categories	Multinomial logit	IIA plausible?
Non-negative integer	Poisson	Test overdispersion
Overdispersed count	Neg. Binomial	Often the safe default
Many zeros	Zero-inflated	Two-process data?
Time until event	Cox PH	Test PH assumption

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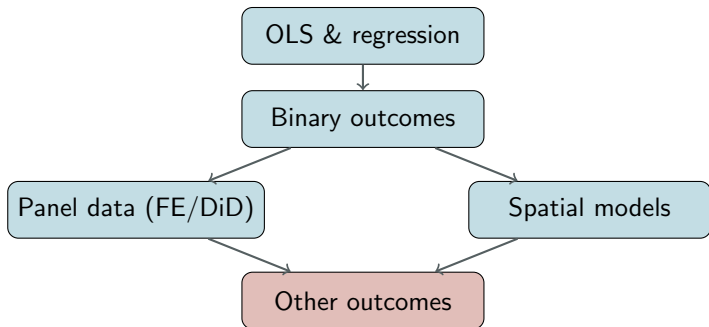
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- **Steps 5–6:** Interpret via AMEs / predicted probabilities (**never raw coefficients alone**); visualize results

This semester



Today: ordinal, nominal, count, duration

(Computing session on April 23rd)

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- Know the relevant **R functions**

Questions?